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(In Italiano)

Progettazione di un'ontologia giuridica per i requisiti di licenza nel diritto del credito: un caso di studio per il controllo della conformità rispetto all'articolo 29 della legge australiana sulla tutela del credito al consumo del 2009

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(In English)

Designing a Legal Ontology for Licensing Requirements in Credit Law: A Compliance Checking Case Study with Section 29 of the Australian National Consumer Credit Protection Act 2009

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Technology Stack

1. Protege
2. OWL 2 DL
3. GRAPH DB
4. SPARQL QUREIS
5. yEd
6. Widoco
7. HTML CSS , JAVASCRIPT
8. GITHUB



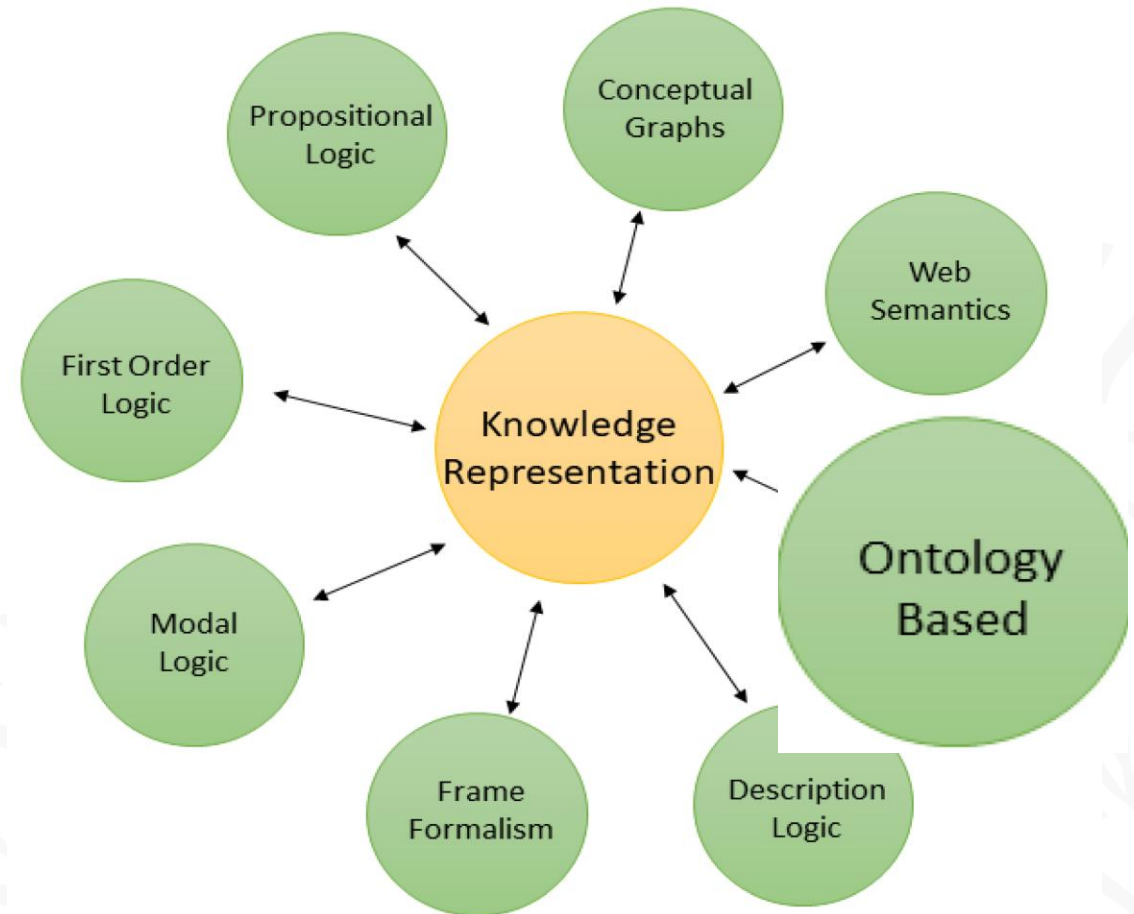
Representation of Knowledge

how we organize, structure, and formalize knowledge so a computer can understand and reason with it.

Data alone= raw facts

Knowledge= data+meaning+relationships+rules

- **KR allows:**
- Reasoning (inferring new facts)
- Consistency checking
- Decision-making
- Sharing knowledge across systems

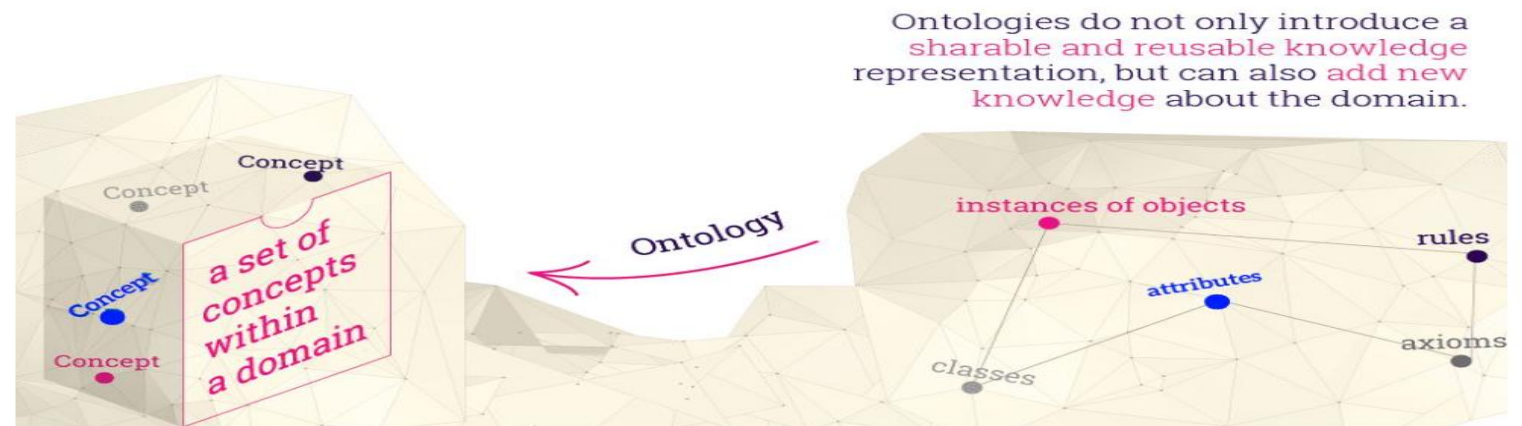


Ontology and why we used it

A formal, structured representation of knowledge that defines **concepts (classes)**, **relationships (properties)**, and **rules (constraints)** within a domain, allowing machines to interpret and reason about it."e.g Set of concepts in a domain

Why?

1. Reasoning
2. Standrized and reusable
- 3.Supports reasoning
4. Ontology Adds Meanings(classes,relationships,Rules/restrictions)
- 5.Legal provisions(capture complexity of law)



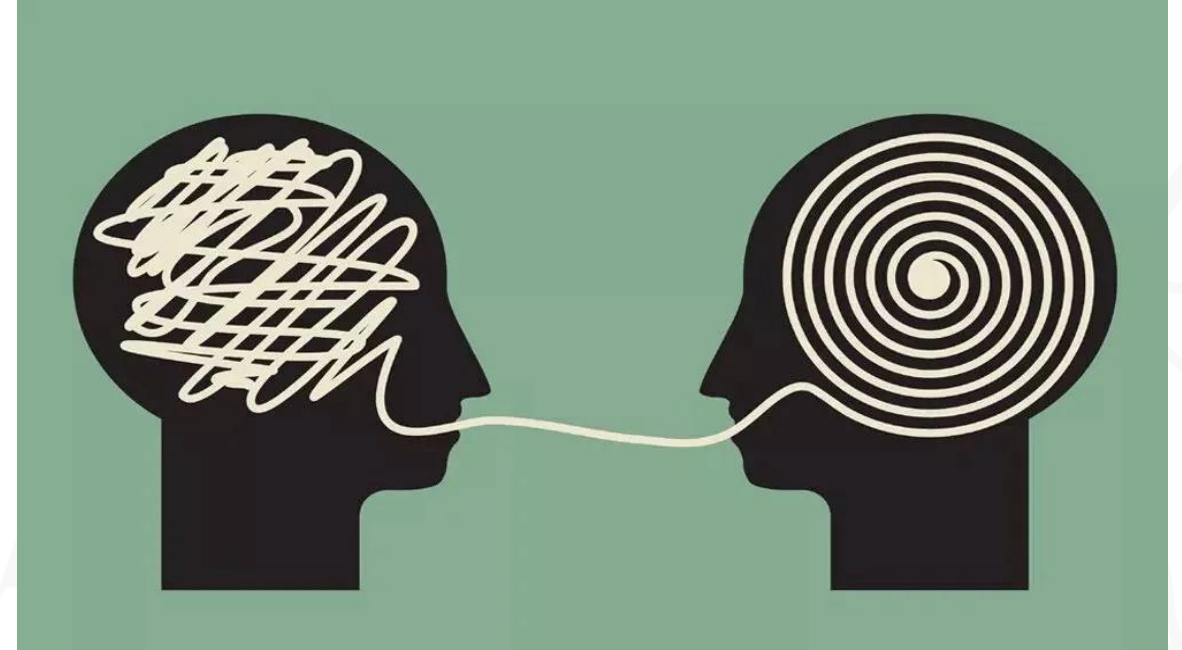
Why Ontology and why not Simpler Options?

Why Ontology?

- Structured and machine-readable knowledge
- Enables automated reasoning
- Reduces ambiguity and ensures consistency
- Uses standards (OWL, RDF) → reusable

Why not simpler options?

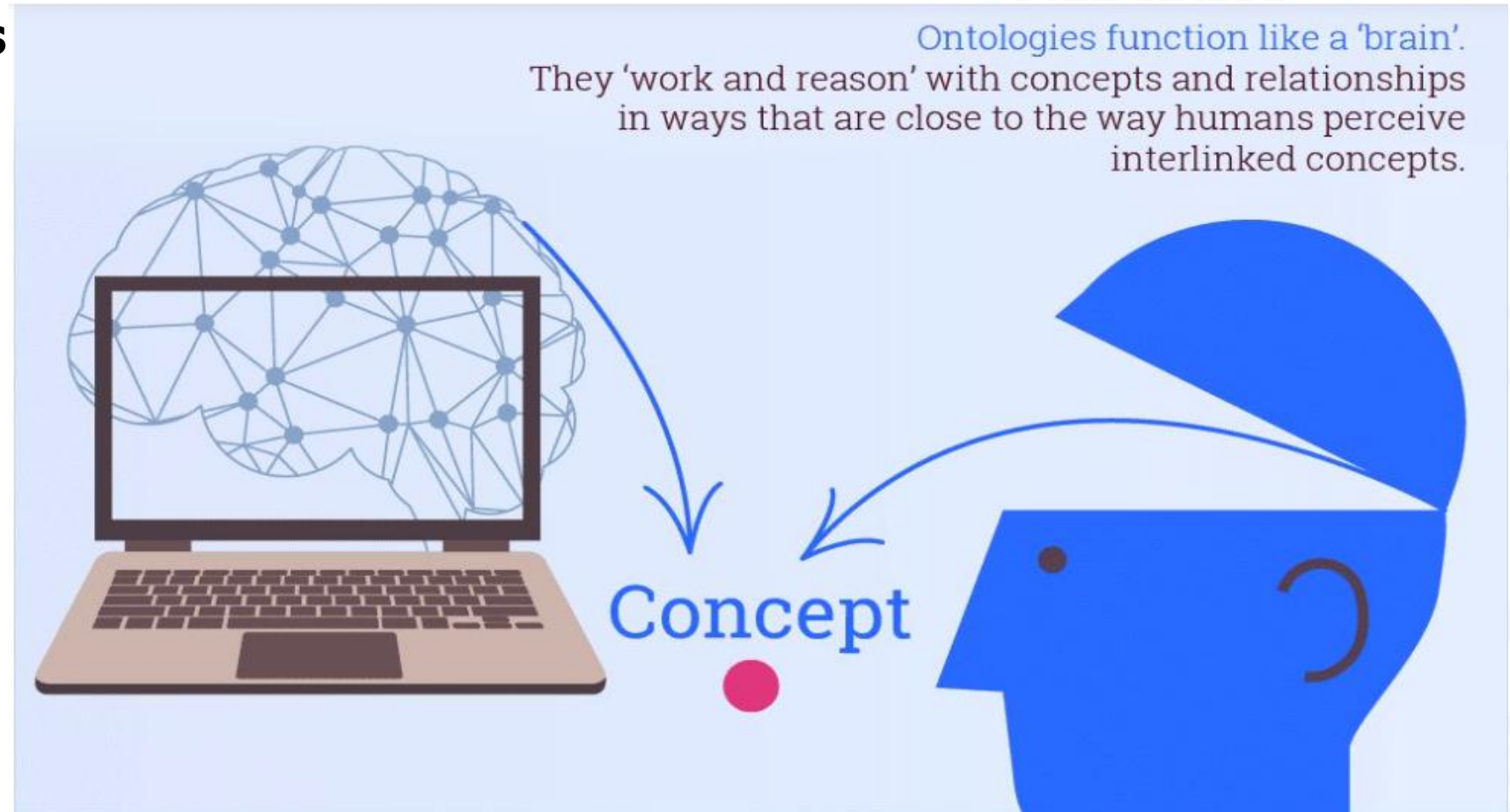
- Databases → Store data but can not reason
 - Spreadsheets → limited structure (no relationship/logic)
 - Text → not machine-readable
- ☞ Ontology enables reasoning, unlike simpler tools.



Reasoning is the ability of a system to automatically infer new knowledge using defined rules

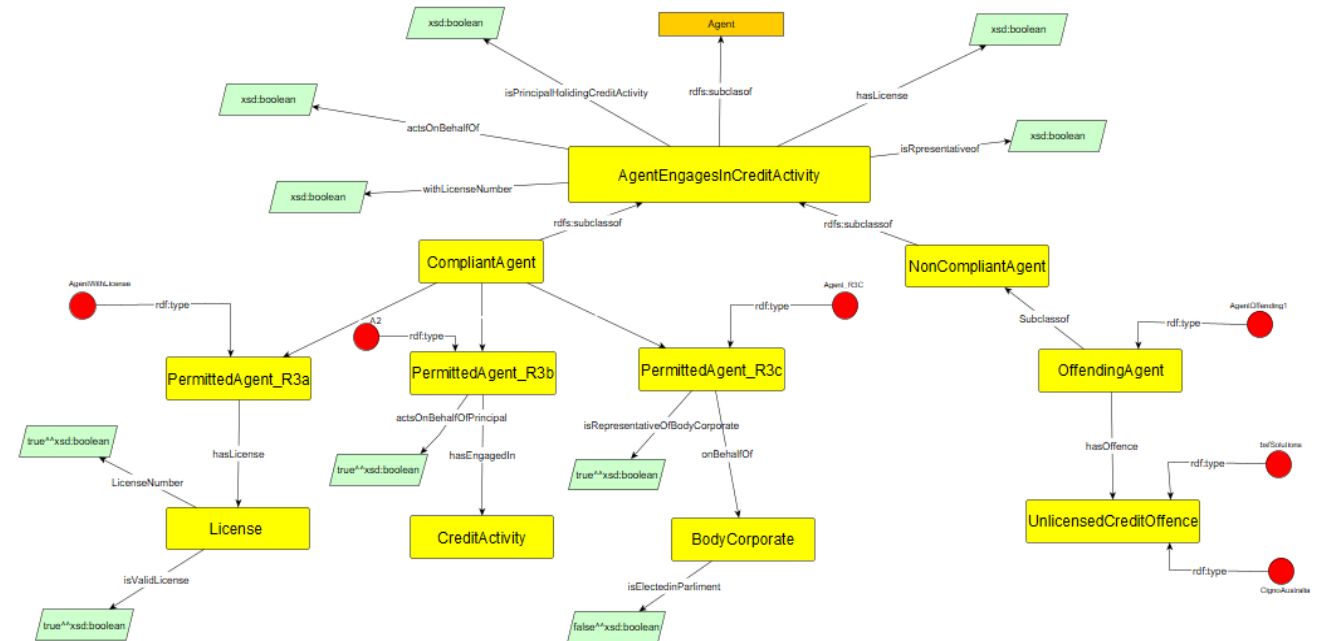
Key Components

- 1.Classes
- 2.Individuals
- 3.Properties
 - a)Object property
 - b)Data property
- 4.Hierarchies



Ontology Visualization Using yEd

1. Provides a **graphical representation** of ontology structure and relationships
2. Hierarchy of Legal Concepts
3. Relationship between entities
4. Compliance/non-Compliance scenarios
5. Improve readability and Understanding

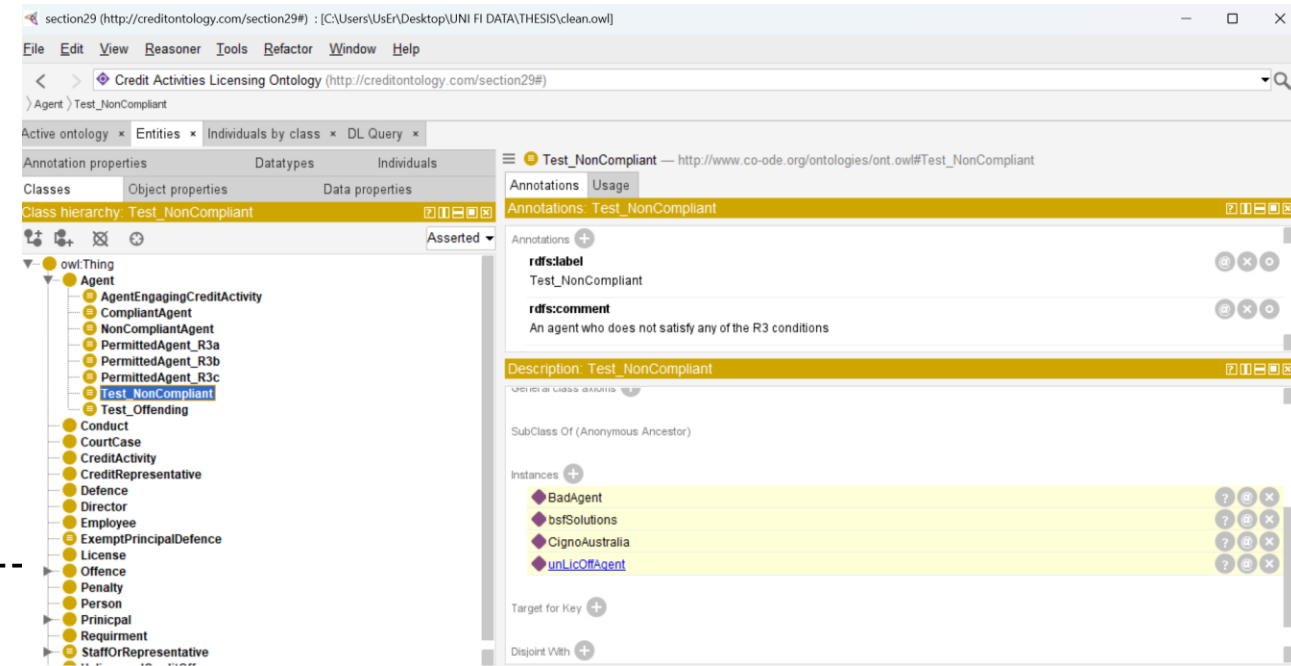


Study Case Section 29

Protege:

Non Compliant Results ,are inferring properly

A ontology example of the Case study



In Grapgh DB

we run a run sparql queries to checking the

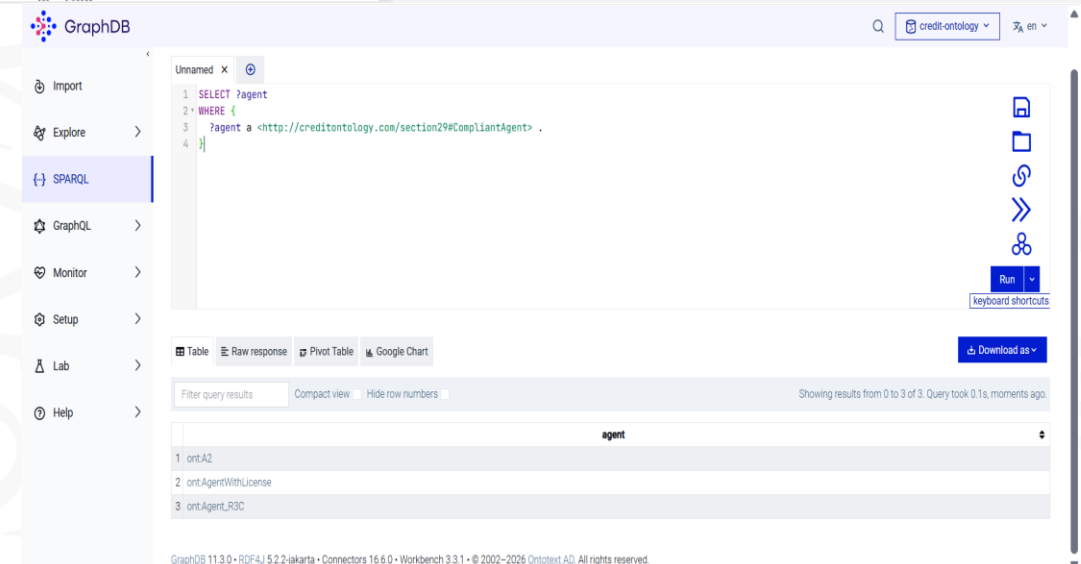
Climpliant agents are inferring properly

SELECT ?agent

WHERE {

?agent a <http://creditontology.com/section29#CompliantAgent> .

}



Ontology Documentation Using Widoco

1. Automatically generates **human-readable documentation** from OWL ontology
2. Improves Understanding and transparency
3. Supports Reuse and extension of technology
4. Ensures **standardized and structured documentation**
5. Output:
 - Interactive **HTML documentation website**
 - Easy navigation of ontology components

UI interface of implementation

Develop by a Website UI using

1)HTML ,CSS,JAVASCRIPT , and Host it in

the github pages

2:provides

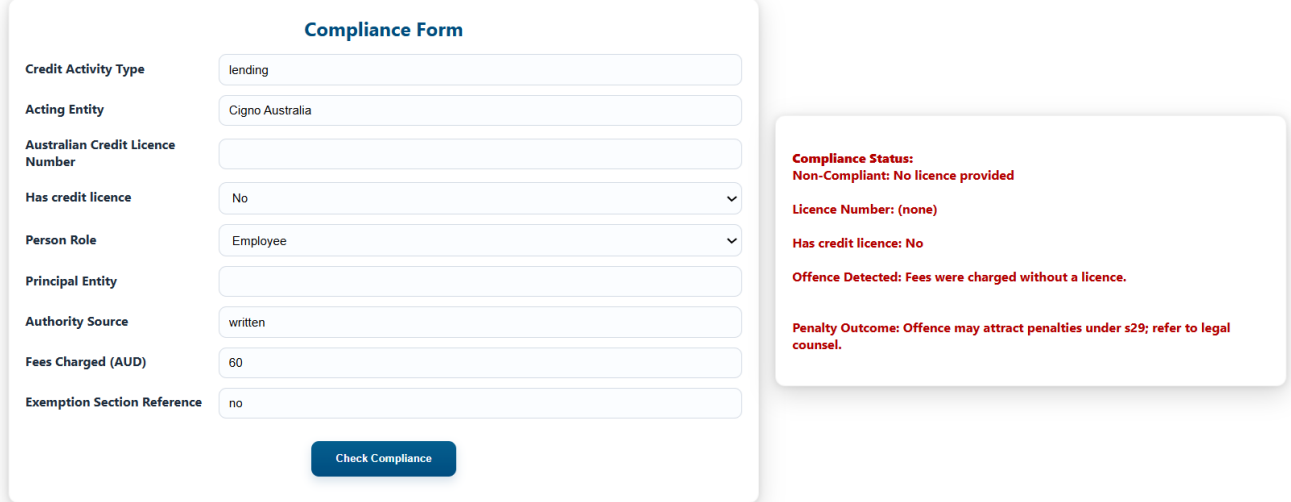
a simple and accesible framework for non-technical users

3. User Input data -> javascript evaluates conditions

4.) display instant results

5.) bridges gap b/w complex ontology and users

6) User freindly and intituitve design ,Real time feedback



The screenshot displays a web application titled "Credit Licensing Compliance Checker" with a subtitle "Section 29 – National Consumer Credit Protection Act 2009". The main form, labeled "Compliance Form", contains several input fields: "Credit Activity Type" (text input with "lending"), "Acting Entity" (text input with "Cigno Australia"), "Australian Credit Licence Number" (text input), "Has credit licence" (dropdown menu with "No" selected), "Person Role" (dropdown menu with "Employee" selected), "Principal Entity" (text input), "Authority Source" (text input with "written"), "Fees Charged (AUD)" (text input with "60"), and "Exemption Section Reference" (text input with "no"). A blue "Check Compliance" button is positioned at the bottom right of the form. To the right of the form, a white box with a red border displays the results: "Compliance Status: Non-Compliant: No licence provided", "Licence Number: (none)", "Has credit licence: No", "Offence Detected: Fees were charged without a licence.", and "Penalty Outcome: Offence may attract penalties under s29; refer to legal counsel."

Evaluation and Discussion

Effectiveness

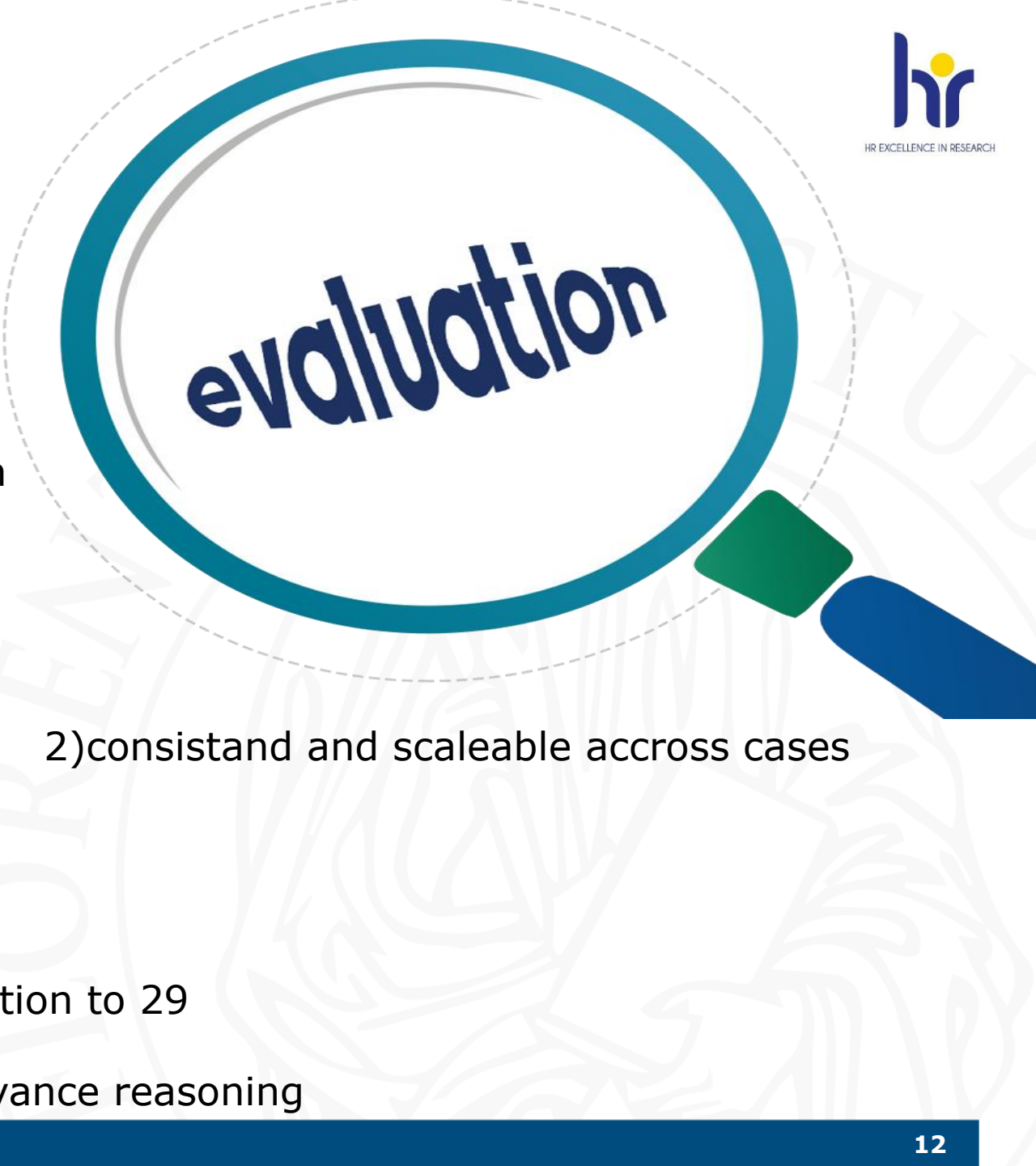
- 1) Accurate Modeling of section 29
- 2) Automatic Compliance classification
- 3) consistent and reliable results

Advantages

- 1) Formal & Structural legal presentation
- 2) consistent and scalable across cases
- 3) built with Protege GraphDB, Sparql

Limitations:

- 1) Simplifies Legal Complexity
- 2) Limited Section to 29
- 3) Depends on data accuracy
- 4) Limited advance reasoning



evaluation

Conclusion

Conclusion

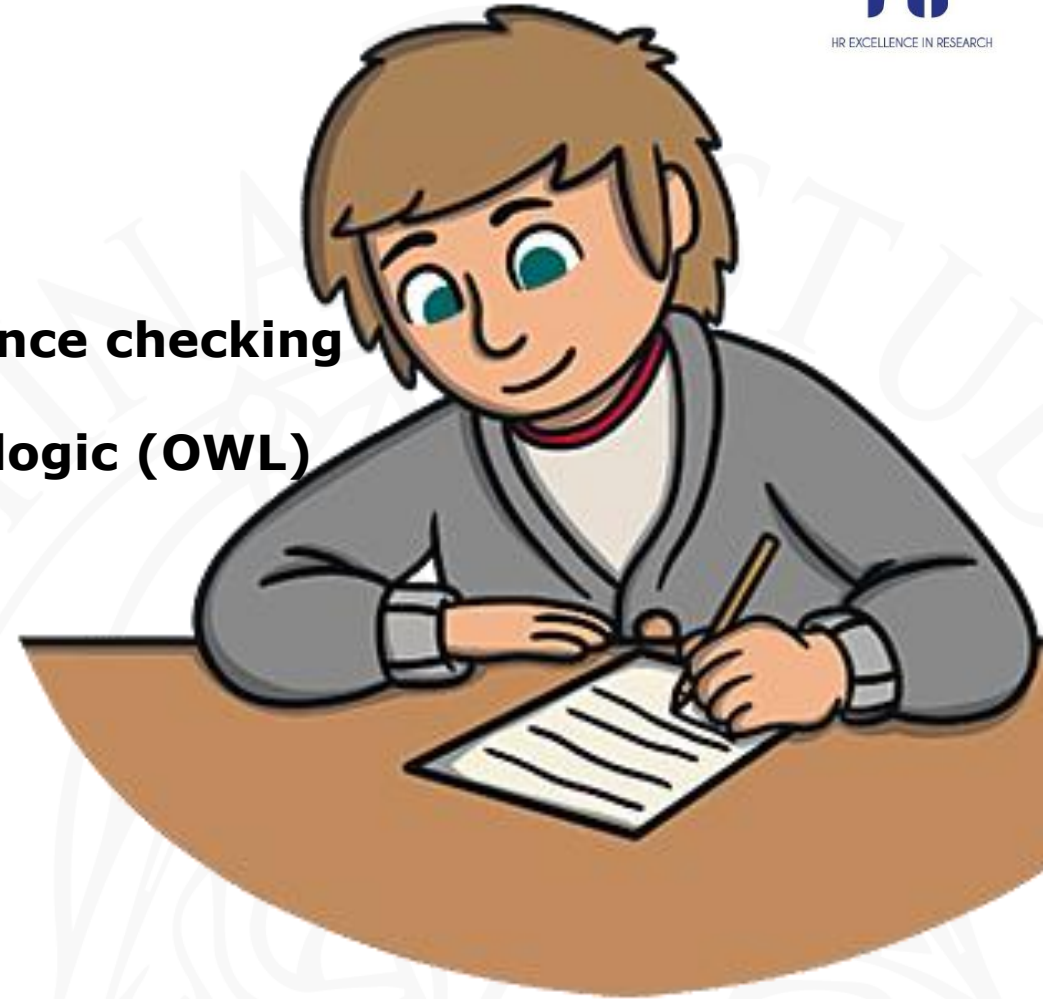
- Ontology for *Section 29* enables **automated compliance checking**
- transformed legal rules into **machine-interpretable logic (OWL)**
- Results are **consistent, accurate, and scalable**

Contribution:

- Connects **legal text** ↔ **computational logic**

Future Work:

- Extend scope beyond Section 29
- Improve reasoning & real-world integration





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Thank you !!

